

Activity 3: Comparing Object Oriented Methodologies

Selecting the Best Development Methodology

1) Booch Method:

Advantages:

Focuses mainly on object oriented design. Encourages modular and reusable code. Suitable for developing complex software systems.

Limitations:

Difficult for beginners to understand. Requires extensive documents. Less standardized compared to UML.

Suitable Application:

Embedded systems, Engineering software, Defense systems, Large object oriented applications

2) OMT (Object Modeling Technique)

Advantages:

Separates the system into object, dynamic and functional models. Easy to understand and analyze. Suitable for transaction based systems.

Suitable Applications:

Banking systems, Inventory management systems, Business information systems, Financial applications

3) UML - Based Development

Advantages:

International standard for object oriented modeling. Supports the complete software development life cycle. Improve communication among developers. Easy to maintain and update.

Limitations:

Requires knowledge of UML diagrams. Large projects may require many diagrams. Documentation may become time consuming

Overall Comparison:

→ Booch Method:

- * Mainly focuses on software design.
- * Best for designing complex object-oriented systems.
- * Less commonly used today because UML has become very industry standard.

→ OMT

- * Mainly focuses on system analysis.
- * Best for analyzing business processes and object interactions.
- * Commonly used for information and banking systems.

→ UML Based Development

- * Covers analysis, design, implementation, testing and deployment.
- * Uses standardized UML diagrams.
- * Most widely used methodology in software industries today.

Conclusion :

Among the three methodologies, UML Based Development is the most widely used because it supports the complete Object Oriented software development life cycle (OOSELC). It provides standardized diagrams that improve communication, simplify software development and makes maintenance easier.